
CYLINDER

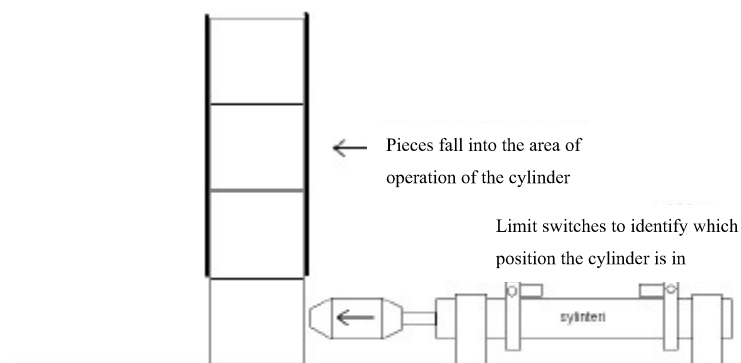
1 INTRODUCTION

The work uses an electrical pneumatics training workstation and a Siemens S7 1200 logic. The equipment used in the exercise is installed in place, and the necessary connections have been made (both electricity and compressed air). The figure shows a workstation ready for making a magazine application.



The purpose of the task is to produce program code that could control the fictitious magazine described in this work instruction.

The magazine has stacked pieces stacked on top of each other, the bottom of which is pushed off by a pneumatic cylinder every three seconds (e.g. on a conveyor belt). Limit switches are attached to the cylinder, with the help of which the logic detects the extreme positions of the piston. The figure shows the imaginary function of the magazine. The operation of the magazine is simulated using the tools found at the workstation (cylinder, limit switches, etc.)

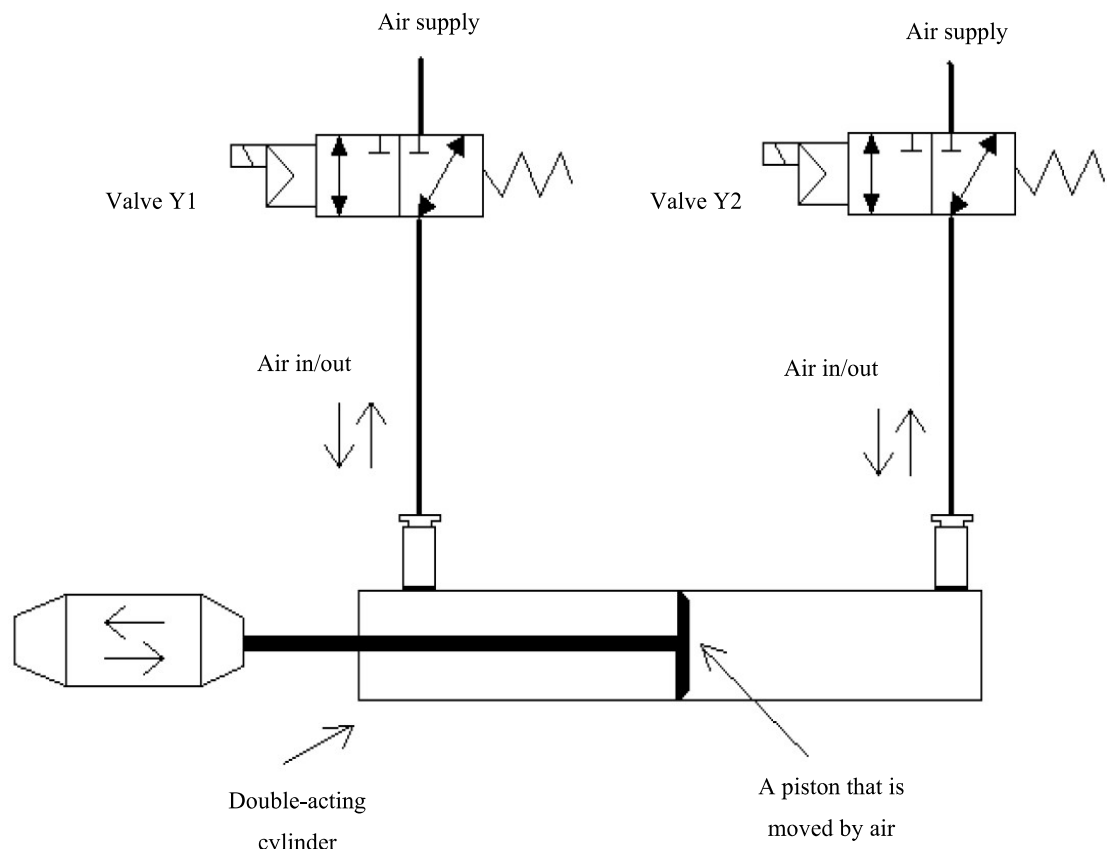


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2 INFORMATION ABOUT THE WORKSTATION

Pneumatic cylinders and valves that control them

A double-acting cylinder is used for this laboratory work. Double-acting here means that the cylinder has two pneumatic fittings, and the direction of movement of the cylinder is determined by which connection the pressure is brought to. The supply of compressed air to the cylinder is controlled by two valves according to Figure 3. The valves are controlled by the binary outputs of programmable logic so that both valves Y1 and Y2 of the "ON/OFF" type are controlled by a separate output channel of programmable logic (Q0.0 controls valve Y2 and Q0.1 control valve Y1). Example of operation: (see Figure 3) when the output controlling valve Y1 is on and the output controlling valve Y2 is off, the cylinder moves in because compressed air passes through valve Y1 to affect the piston in the cylinder, and the air on the other side of the piston can escape through valve Y2 into the outside air.



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Limit switches

The work uses limit switches mounted around the pneumatic cylinder used (see Figure 2). The limit switches magnetically detect the piston moving inside the cylinder. The limit switch gives a signal when the piston enters its operating range.

In this task, limit switches are used as limits of movement of the cylinder, so that one detects when the cylinder is outside and the other when the cylinder is in the inner position. The limit switches are connected to the logic inputs in accordance with Annex 1.

3 SAFETY

The pneumatic cylinder poses a risk of compression. Its movement is quite fast and powerful. Therefore, under no circumstances should you put your hands, fingers, etc. parts of the body to the operating range of the cylinder when pressure is applied to the equipment. At the beginning of the work, find out how to depressurize the equipment as quickly as possible.

At the workstation there is a pressure regulator, which is set to a value of about 2 bar. In no case should the value of the supply pressure be changed without the permission of the staff, since too much working pressure makes the movement of the cylinder dangerously strong. If the pressure is too low, the directional valves used in the work will not work correctly.

CYLINDER**4 TASK**

1. Make a TIA Portal project and add the CPU used in the work. Test loading HW and an empty program into logic. If there are any problems with this, clarify them with the instructor.

2. Make a function (FC1) for which you are making the application. Make a program for FC1 that implements the functions described below (the I/O address table is given in Annex 1, load the program into logic and test the operation of the program in monitoring mode as necessary):

- The application is started with a reversible push-button (Start) with an indicator light on (control is obtained from the logic binary output) when the application is on
- a warning flash of about 6 seconds (for this purpose, make a circuit that produces a blinking rate frequency e.g. 1 Hz) before the cylinder starts working when the application is started
- The cylinder pushes the lowest piece out of the stack
- After pushing, the cylinder returns to the inner position and waits for three seconds, after which again pushes out the next piece
- The cylinder continues to operate in accordance with the two previous points until the application is stopped
- The application is stopped by the reversible pushbutton (Stop). When the application is stopped, the cylinder should return to the "inner position" and remain there.
- The position inside the cylinder is determined by the limit switch installed on top of the cylinder
- The extreme position of the cylinder when pushed out is identified by another limit switch installed on top of the cylinder.

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When the job is finished, you can load it into logic and test (You can also download the program to step and test individual functions). You can transfer the program (OB1 and FC1) using the method you learned in the class or, alternatively, in the following way: activate OB1 and FC1 → on top of them, press the right mouse button to open a menu → select the command DOWNLOAD found in the menu under PLC. If the program reports that the logic is already OB1/FC, and asks if you write yes over them. Turn on the pressure at the workstation before testing. If necessary, ask the staff for instructions on how to turn on the pressure when the program starts to be ready. After testing the program, present it to the teacher (introduce the activity and present the program and symbol board that you have made).

ADDITIONAL TASK: Done if there is time left. Change the application so that after pressing the start button, the cylinder performs 5 pushes, after which the application stops to wait for restarting. The Stop button should pause the operation as in the original application. Present the finished work to the instructor.

ADDITIONAL TASK 2: Add to the application a function that returns the cylinder to the inner position of the piston and stops the program if the piston does not reach the outer limit within 5 seconds of steering.

CYLINDER**5 ATTACHMENTS****Annex 1. I/O's addresses**

ADDRESS	EXPLANATION
I0.0	"Start" -push-button input
I0.1	"Stop" -push-button input
I0.2	Input of the "out position" limit switch (mode = '1' when cylinder is in the out position)
I0.3	Input of the "In position" limit switch (mode = '1' when cylinder is in the in position)
Q0.0	Controls valve Y2, which directs the cylinder to the out position (out)
Q0.1	Controls valve Y1, which directs the cylinder to the in position (in)
Q0.2	Output controlling the warning indicator (blinking at the beginning).
Q0.3	Led output (indicates whether the app is on)